Megatron-FSDP / HSDP with EP

Shard every byte of model state across data parallelism

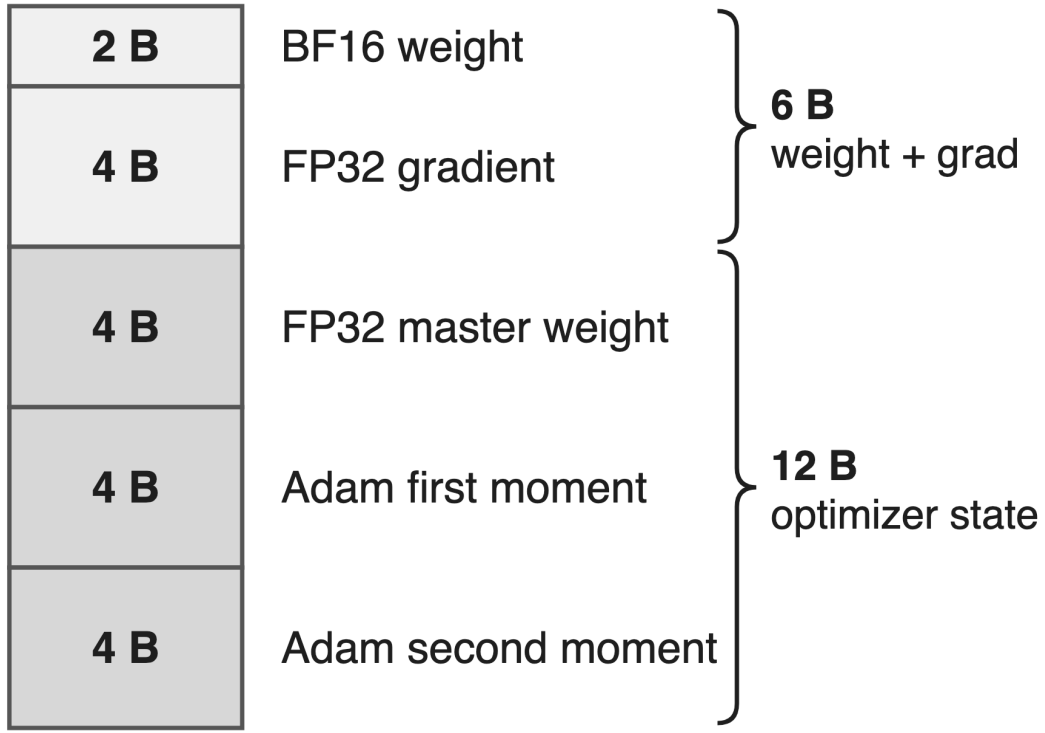
NVIDIA · Megatron-Core (HSDP for MoE v0.13, FSDP + EP v0.15)

Explainer deck · Industrial Pretraining Architecture & Infra Tech Landscape (2026)

Parallelism Strategy · github.com/NVIDIA/Megatron-LM

Giant MoEs keep every expert's state resident

- DeepSeek-V3: **685B** params, **37B** active (**18×** gap)
- Every parameter carries **18 B** of state
- Plain data parallelism copies all of it to every rank



Model state alone nearly fills an H100

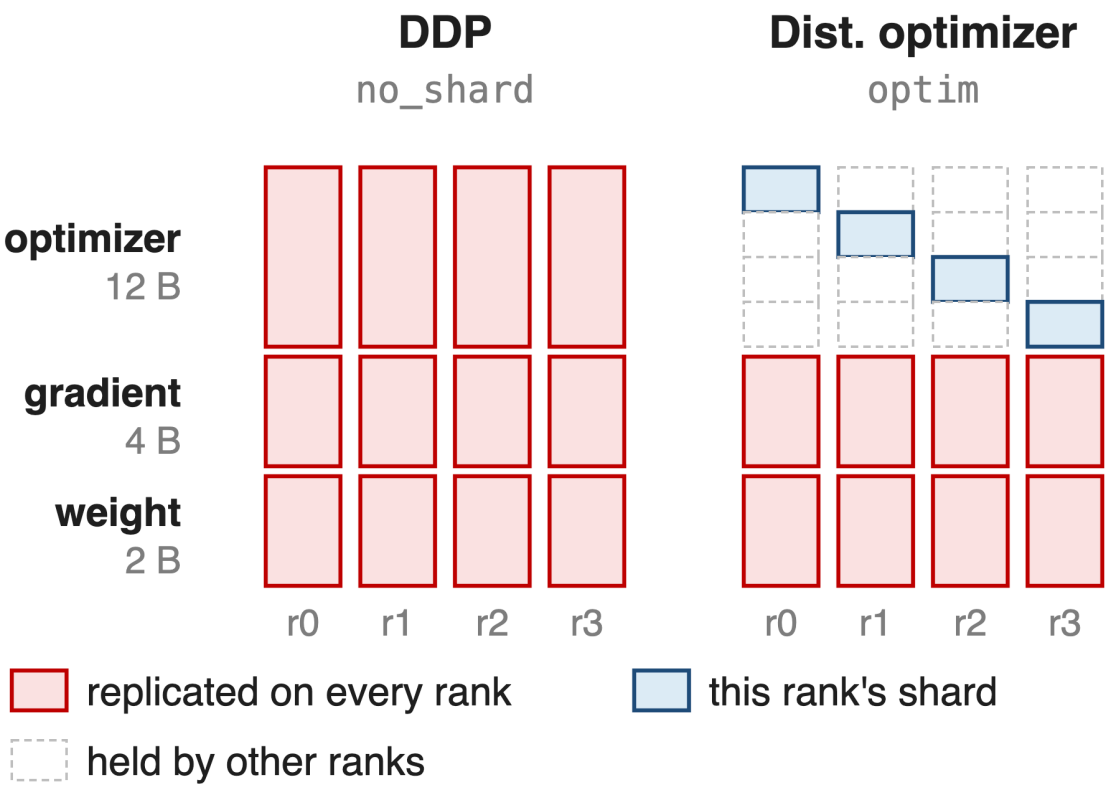
68.5 GB

weights, gradients and optimizer state per GPU: DeepSeek-V3, BF16, 256 GPUs; activations add 131 GB; an H100 has 80 GB

arXiv 2603.07685, Table 3 (PP4 × VPP4 × EP64): 36.4 GB + 32.1 GB

The distributed optimizer stops at 6 bytes

- ZeRO-1 shards only the 12 B of FP32 state
- Weight + grad stay: **6 + 12/d** B
- **More PP / EP / TP**: bubbles, stage balancing, cross-node all-to-all



Insight: a rank needs each layer only briefly

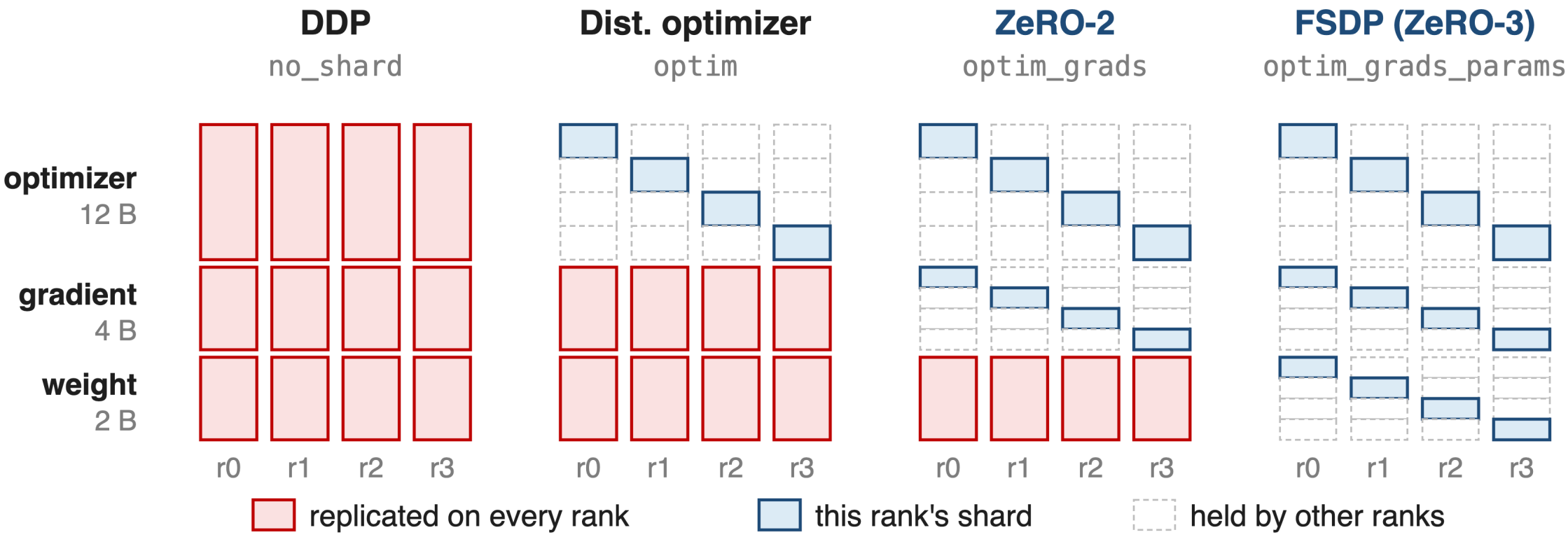
- DP replicas hold identical model state
- Each rank computes one layer at a time

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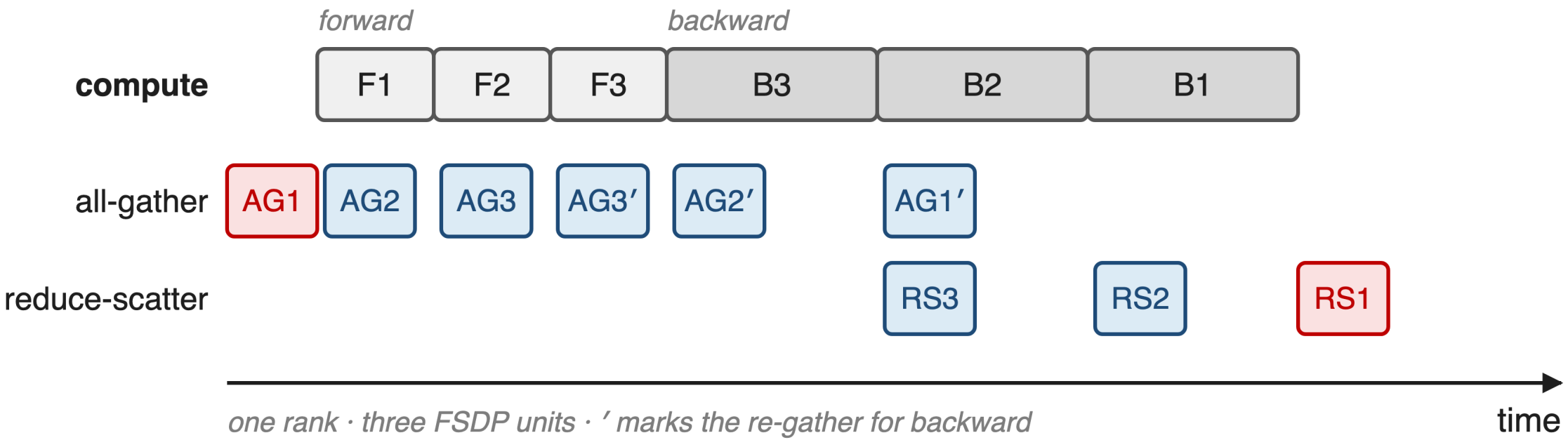
Keep 1/d of everything; gather the next layer while this one computes. DP size becomes the memory knob

One flag picks how much state to shard



--use-megatron-fsdp --data-parallel-sharding-strategy {no_shard, optim, optim_grads, optim_grads_params}; checkpoints use --ckpt-format fsdp_dtensor

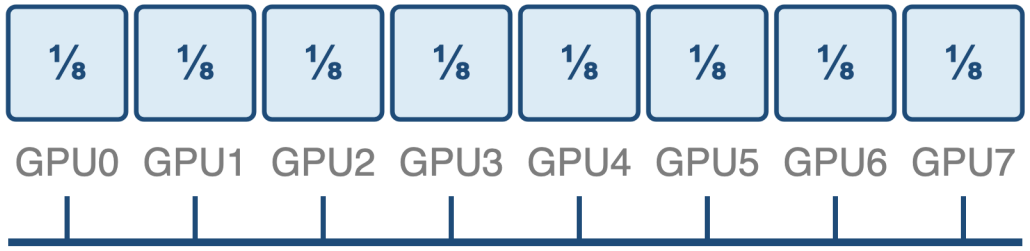
❶ Gather each layer just in time



Two persistent buffers alternate between units, so at most two units are ever un-sharded

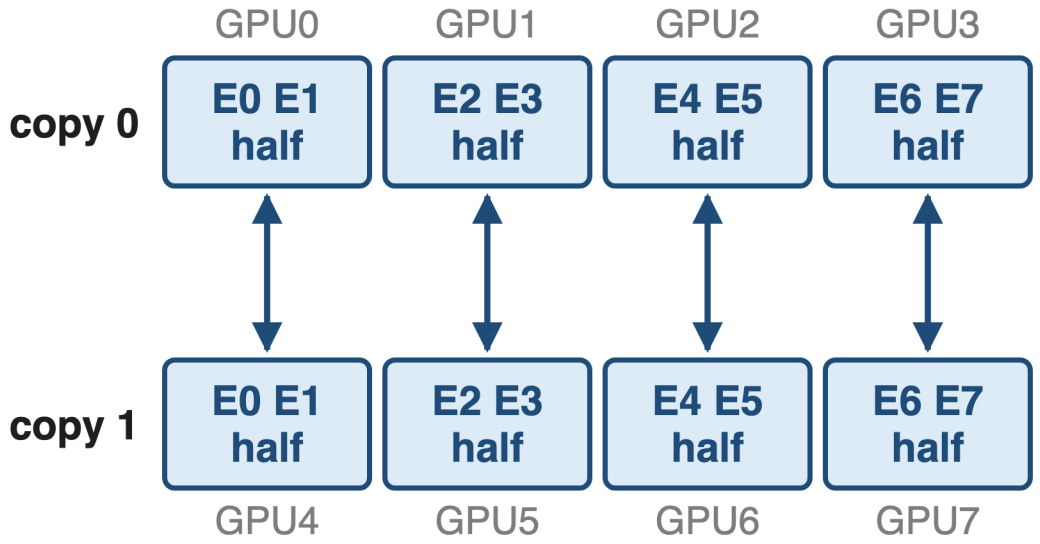
❷ Experts shard over their own replicas

(a) Dense mesh: attention and norms



AG and RS over all 8 ranks

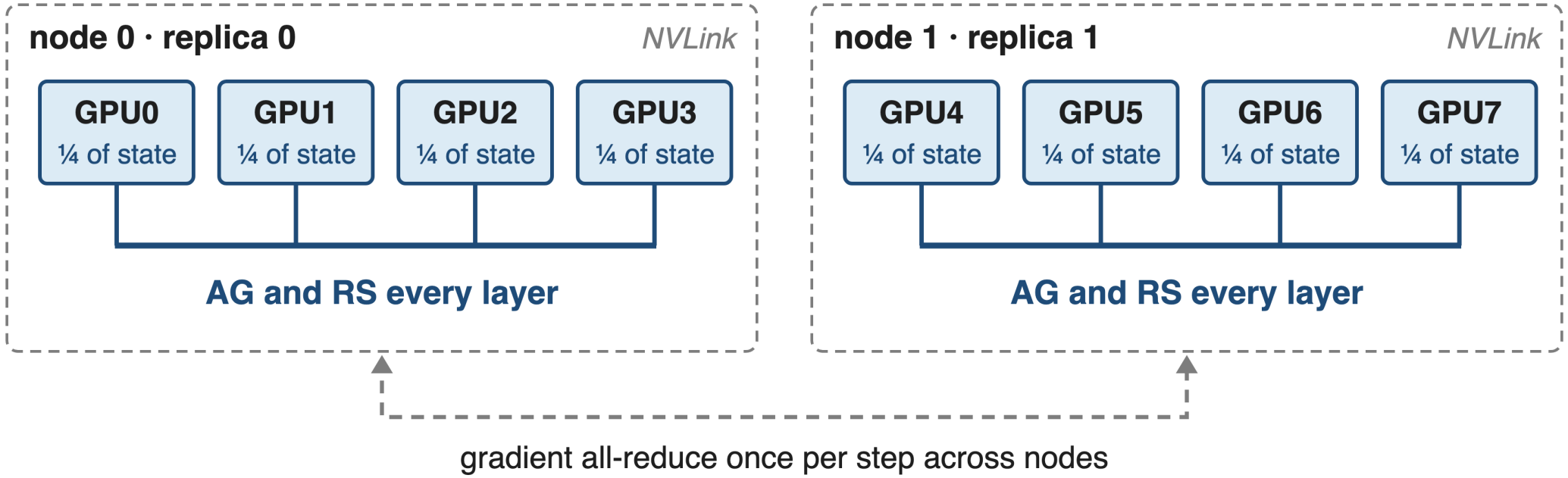
(b) Expert mesh: EP 4 × EDP 2



AG and RS within each column: 2 ranks

Dual DeviceMesh: dense modules shard over DP, experts over EDP, so expert collectives stay small

❸ HSDP keeps per-layer traffic inside NVLink

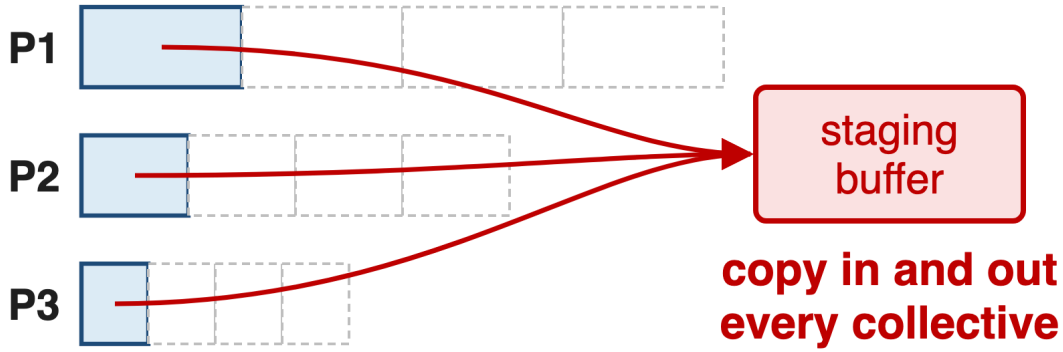


--num-distributed-optimizer-instances 2 sets the outer size; --outer-dp-sharding-strategy no_shard (HSDP) or optim (HFSDP)

HFSDP also shards optimizer state across nodes and syncs it once per step

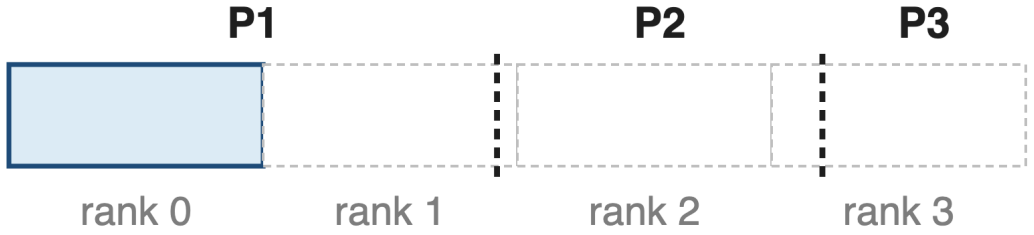
❹ Flat buffers make collectives zero-copy

(a) Per-parameter shards (FSDP2)

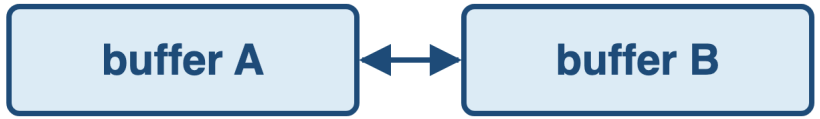


rank 0 owns a quarter of each parameter

(b) Flat per-module buffer



AG and RS run in place on NCCL-registered memory

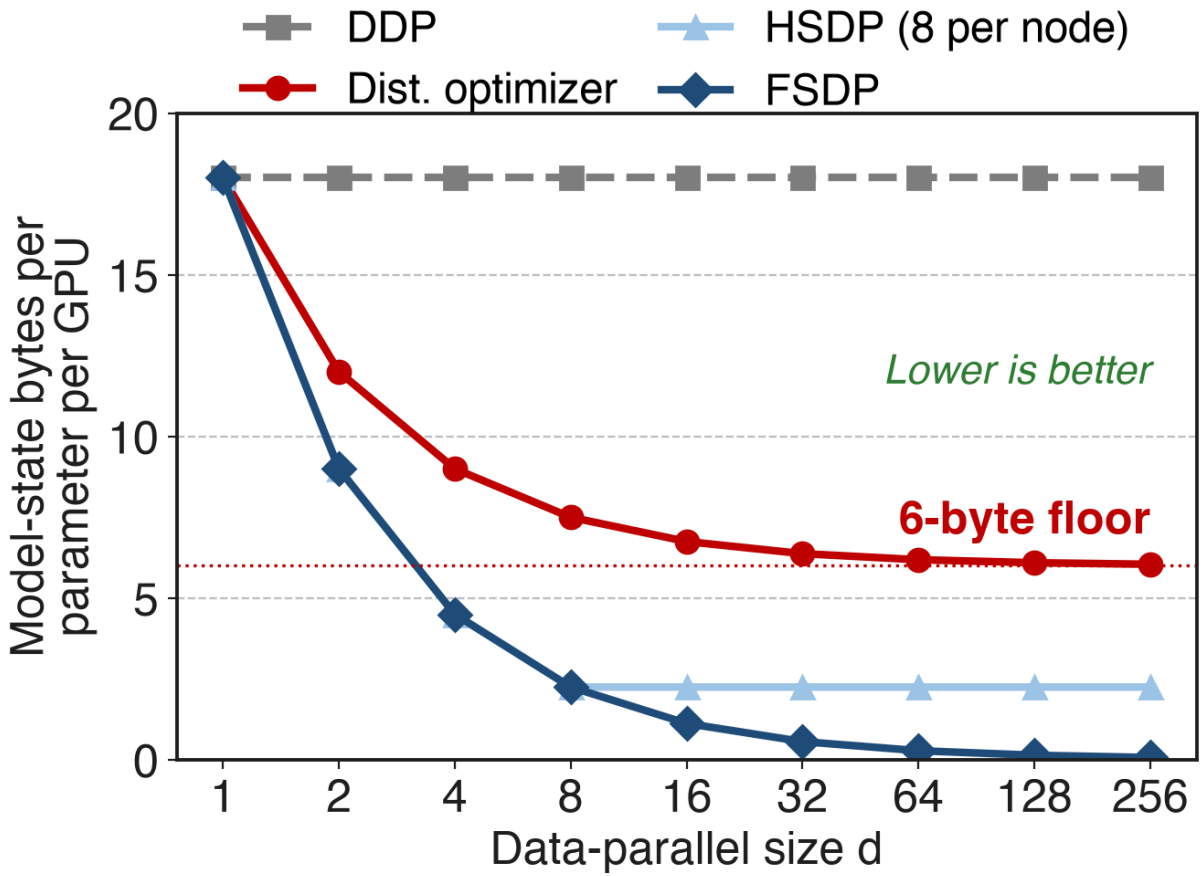


units alternate between two buffers

~10% less comm overhead on Llama3 405B; comm kernels drop from 8–32 to 1–4 SMs on NVLink

Full sharding removes the 6-byte floor

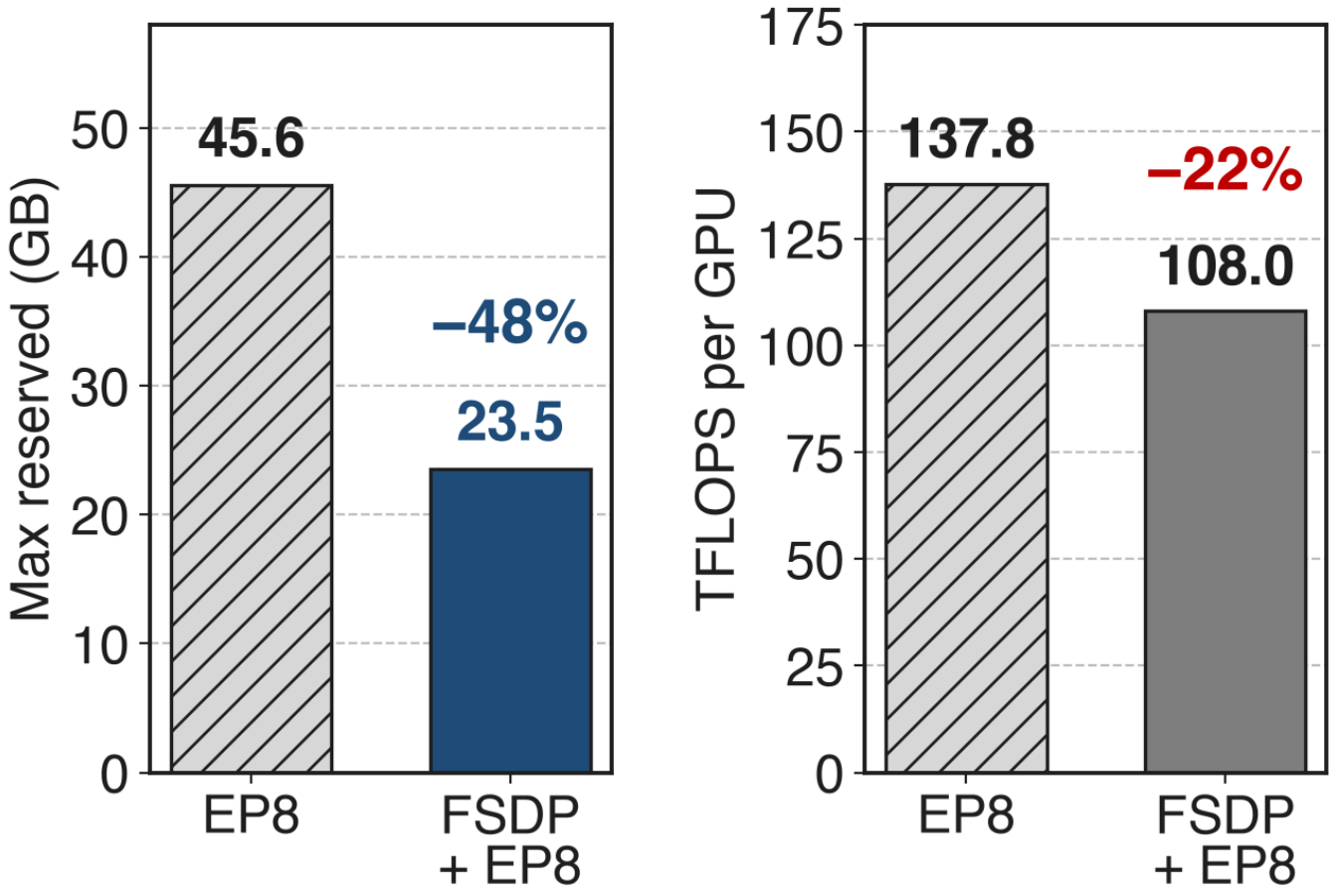
- At d = 64: **6.19 B** vs **0.28 B** per parameter
- HSDP at 8 per node: **2.25 B**
- Memory falls with every rank you add



Analytic: BF16 weight, FP32 grad, FP32 Adam; excludes two transient units and activations

FSDP + EP halves memory on Qwen3-30B-A3B

- 32 H800, user report (Megatron-LM #2772)
- **–48%** peak reserved memory
- **–22%** TFLOPS: worth it only when memory binds



DeepSeek-V3 recipe drops pipeline parallelism

- **N-D parallel (MCore tuned)**
- H100: TP2 · PP8 · EP64 · VPP4
- **Stage balancing**, MTP and output-layer placement

- **Megatron-FSDP script**
- TP1 · EP8 · no PP · ZeRO-3
- Tune two knobs: EP size, sharding degree

The cost: 1.5× traffic that must hide

- ZeRO-3 moves **1.5×** plain DP's bytes
- Hiding it needs compute: **bigger micro-batches**
- **Activations not sharded**
- fsdp_dtensor checkpoints; convert from torch_dist
- Double buffers cost memory, want uniform layers
- No Dynamic CP; Torch FSDP2 path experimental

Summary

- DeepSeek-V3: **68.5 GB/GPU** of model state; the distributed optimizer floors at **6 B** per parameter
- **Insight**: a rank needs each layer only briefly, so shard it and gather just in time
- Dual mesh for experts, HSDP inside NVLink, zero-copy buffers
- **18/d** bytes per parameter; **–48%** memory on Qwen3-30B-A3B

https://github.com/NVIDIA/Megatron-LM